

Machine Learning Model Evaluation through Diabetes Prediction

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Abstract

Diabetes mellitus is a rapidly growing global health concern affecting over 537 million individuals, projected to exceed 783 million by 2045. This project investigates machine learning-based classification models for predicting diabetic status using clinical, biochemical, demographic, and behavioral health features. The objective is to develop a multi-class classification system capable of identifying patients as healthy, pre-diabetic, or diabetic with high predictive reliability and interpretability. Using models such as Logistic Regression, Random Forest, K-Nearest Neighbor (KNN), and two Multi-Layer Perceptron (MLP) architectures, we evaluate performance using accuracy, F1-score, precision, recall, and AUC metrics, with special consideration of dataset imbalance. The results demonstrate that the MLP architecture yields the strongest predictive performance across the three categories, with an accuracy of 85% and AUC of 79%, with our more advanced MLP showing an improving pre-diabetes recall, which is an outcome vital for preventative healthcare. This work highlights the potential of machine learning to support medical decision-making and improve early identification of diabetes risk.

1. Introduction

Diabetes mellitus is a chronic metabolic disorder characterized by impaired glucose regulation and long-term organ damage. With more than 537 million people currently living with diabetes and projections estimating up to 783 million cases by 2045, early detection and risk forecasting have become increasingly indispensable to global public health. Type-2 diabetes, in particular, can often be managed or reversed if detected early which underscores the need for predictive tools capable of identifying high-risk individuals before irreversible complications emerge.

Machine learning provides a powerful framework for clinical risk prediction, offering the ability to analyze high-dimensional medical data and uncover subtle patterns not easily identifiable through traditional diagnostics. The goal of this project is to develop and evaluate machine learning models that classify patients as diabetic, pre-diabetic, or healthy based on key physiological, biochemical, and demographic indicators. The dataset used in this study includes features such as age, BMI, blood glucose, cholesterol profile, urea, creatinine levels, lifestyle factors, and biological markers such as HbA1C. These attributes serve as valuable predictors for diabetes development and severity progression.

To achieve this objective, we implemented several supervised classification models, including Logistic Regression, Random Forest, K-Nearest Neighbor (KNN), and Multi-Layer Perceptron (MLP) networks. Particular attention was given to the imbalanced class distribution, where healthy individuals significantly outnumber pre-diabetic and diabetic cases. Accuracy alone would therefore misrepresent model quality, so performance was measured using macro-F1, AUC-ROC, and per-class recall, with special focus on improving prediction consistency for the minority pre-diabetes class. Our strongest-performing model, a class-weighted MLP architecture, demonstrated more balance among the three classes, but still suffered from some imbalance, which we will discuss later, suggesting that neural networks hold substantial promise for early diabetes risk screening.

2. Related Works

Several prior works have demonstrated the power of machine-learning classifiers to predict diabetes status using clinical and demographic data. For instance, Classification and prediction of diabetes disease using machine learning paradigm used a dataset of 6,561 respondents from the National Health and Nutrition Examination Survey (NHANES), applied logistic-regression for risk-factor selection (identifying age, BMI, blood pressure and cholesterol among others), and then built a random-forest classifier — achieving 94.25% accuracy and an AUC of 0.95 under 10-fold cross-validation [1]. More recently, Prediction of Type 2 Diabetes Based on Machine Learning Algorithm applied a

multiclass classification approach (normal, prediabetes, and diabetes) on longitudinal electronic-health-record data, using features such as fasting glucose, HbA1c, BMI, lipids, and lifestyle indicators, then comparing Logistic Regression, Random Forest, SVM, and ensemble models to identify at-risk individuals before disease onset. [2] These studies support the plausibility of using routine clinical and biochemical variables for diabetes risk stratification, and provide methodological templates (feature selection, cross-validation, class balancing) which closely align with and substantiate the strategy adopted in our project.

3. Proposed Method

We propose to evaluate our dataset on five different machine learning models, first a Logistic Regression model, which serves as our baseline model, a Random Forest model, a K-Nearest Neighbor(KNN) model, and lastly 2 Multi-Layer Perceptron(MLP) models with differing architectures, to see which model performs the best overall on this dataset.

Let's dive into the details on how we implemented our models. Firstly, for our Logistic Regression model, we used Python's SkLearn library to provide us with the implementation of the Logistic Regression model, some important parameters we tuned our model to was the number of max iterations, we set it to 1000, because as you know our dataset is quite large, we wanted our model to learn the data and its underlying patterns and relationships well. Moving on to our next model, our RandomForestClassifier also utilized the

same SkLearn Python library for its implementation. An important parameter for our model was the number of estimators, which is the number of decision trees in our “forest.” We kept this at the default value of 100, as this is a good starting point for a dataset of 200k data instances. Next, for our KNN model, we utilized Python’s SkLearn library for this model’s implementation as well. An important parameter to note for our model was the number of neighbors, we set it to 6, which is a decent starting point for a dataset of 240k examples.

For our MLP implementations, we implemented two differing architectures to see which one will perform better overall on our dataset, one of our implementations was more advanced and was more suited for a dataset of this kind. For our less advanced MLP we utilized SkLearn’s MLPClassifier module, we started with an input layer of 21 features, keep in mind this is the 21 other features in our dataset instances, that contribute to our class label, we utilized all features, even the ones that are not as significant. Our first hidden layer had 128 neurons, with ReLU activation, then the next one had 64 neurons with the same activation, with our last hidden layer having 32 neurons with ReLU activation. Our output layer had 3 neurons, because we had 3 classes, with softmax for multi-class classification. Some other important things to note about this model are that we used an Adam optimizer, which is overall a very solid choice especially for larger datasets. We also had an adaptive learning rate while training our MLP, as this is the standard with an Adam optimizer. For our loss we had had

an L2 loss with an alpha of 0.0001. Another important thing to note is we had early stopping so we can save our computational resources. We ran this specific model for 10 epochs.

For our more advanced MLP model, we used the well-known Python library Keras for our MLP implementation. Just like our less advanced MLP implementation we used all 21 attributes for our input dimensions, also we had 3 output features. After our input layer with 21 attributes, we had a hidden layer with 256 neurons, we used a SeLU activation, with a kernel_initializer of “lecun_normal”, which is the standard kernel initialization. We used a SeLU activation function because it helps mitigate the challenges we face in a very imbalanced class dataset by using a small negative slope for negative input values. This prevents the severe suppression of minority classes which can happen when you use activation functions like ReLU. So overall, with this model we want the minority class to actually contribute to the training of the model. After this hidden layer, we had an AlphaDropout layer of 0.1. This was also used instead of regular dropout because it is obviously more compatible with the SeLU activation function, and it's generally better for large class imbalance datasets, such as the one we are dealing with. Moving on to the next hidden layer we had 128 neurons with the same parameters as the previous hidden layer, such as kernel initialization, and the activation function. Then we had an AlphaDropout layer of 0.1. Then after that we had another hidden layer with 54 neurons, other parameters remaining the

same. Then after that we had a AlphaDropout layer of 0.1, then finally we had an output layer with the output dimensions equaling to three, with the standard softmax for multi-class activation. We planned to train our model for 50 epochs, with early stopping enabled. Another extremely important point about this far more advanced model is that we actually applied weights to each class, applying the highest weight to the minority class, which happened to be the pre-diabetes class. We assigned each class with a weight using this formula: $\text{Weight} = (\text{Total Samples}) / (\text{Number of Classes} \times \text{Class Count})$. As you can imagine, classes with a lot of data instances had a smaller weight. Using this formula, we ended up with the following weights: {0: 0.396, 1: 18.26, 2: 2.39}, class 0 being healthy, class 1 being pre-diabetes(highest weight class), class 2 being diabetes. So our model trained on our data given these class weights, so it gave a higher importance to our minority class. Lets see how all of these machine learning models perform next.

4. Experiments

The data set we used for this experiment was from Centers for Disease Control and Prevention's (CDC) called Behavioral Risk Factor Surveillance System BRFSS for the year 2015. This Dataset contains almost 254,000 patient records with many features such as clinical, demographic and behavioral health indicators. The target variable classifies the patients into three categories; 0: For No diabetes 1: For Pre-Diabetes and 2: For Diabetes. Key features included are cholesterol, BMI, high blood pressure,

smoking status, physical activity, fruit and vegetable consumption and other important factors such as age, sex, physical activity. An important thing to note is that this dataset is highly imbalanced; it had the vast majority of its data instances in either class 0, or class 2, with only 4613 data entries! The other two classes were also highly imbalanced, over 200k entries were data instances classified as healthy, and only 35k classified as diabetes. This proved to be an issue, as we believe a more aggressive class balancing technique should've been implemented.

4.1 Experimental Setup

All the experiments were conducted using Python 3.9 with scikit-learn for the traditional machine learning models and Tensorflow/keras were used for neural network implementation. For the MLP training the data was divided by the help of stratified sampling 80% for the training and 20% for training maintaining about 80-20 ratio: 202,944 : 50,736. For the second MLP the dataset split used was a three way 70% training and 15% for validation and 15% for testing. To ensure all the features applied contributed equally for the training we used Standard Scalar normalization. For the imbalance in the dataset we used class weights which penalized the misclassification of minority classes during training.

4.2 Evaluation metrics

Given the class imbalance, using alone accuracy to assess the model would be insufficient, so we applied multiple evaluation metrics: such as Accuracy,

Precision, Recall, F-1 Score and AUC-ROC-Area under the Receiver Operating Characteristic Curve. Special importance was placed for per class recall, specially for pre-diabetes as missing the pre-diabetics patients would cause cost the most in a clinical screening context. Also it's important to also pay attention to the diabetes recall, as this class was pretty sparse too, although way less than the pre-diabetes class.

5. Results and Discussion

TABLE I

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.8370	0.78	0.84	0.79	0.8138
Random Forest	0.8428	0.80	0.84	0.81	0.8200
KNN (k=6)	0.8250	0.76	0.82	0.77	0.7329
MLP (128-64-32)	0.8487	0.80	0.85	0.81	0.7843
MLP (Class-Weighted)	0.6213	0.85	0.51	0.41	0.7652

This table represents the performance for all the evaluated models. Assessing this table we can clearly see that the MLP archives the highest overall accuracy at about 84.87% which was closely followed by Random forest which achieved 84.28%. However these results are achieved because of correctly classifying the majority class while hugely ignoring the minority class. It's only when we take a closer look into the per-class

recall, we can really see which model is more clinically useful.

5.1 Per-Class Performance Analysis

TABLE II

Per-Class Recall Comparison

Model	Healthy (0)	Pre-diabetes (1)	Diabetes (2)
Logistic Regression	0.98	0.00	0.15
Random Forest	0.97	0.00	0.20
MLP (128-64-32)	0.98	0.00	0.15
MLP (Class-Weighted)	0.62	0.26	0.65

This table is based on Per-class Recall analysis. This table reveals the impact of class imbalance. It can be seen that Logistic Regression, Random Forest and standard MLP achieved near to perfect recall for the healthy class/No-Diabetes (from 97%-98%) but completely failed to identify the Pre-diabetics class, also the Diabetes class was not much better. These models tend to classify everything towards the majority class leaving behind the minorities which results in no clinical utility for early diabetes screening.

The class-weighted MLP on the other hand presents a completely different result. By applying class weights and using a self-normalizing architecture alongside SeLU Activation, this model helps achieve 26% recall for the class 0 : Pre-diabetics.

Another thing to note is that it achieves a 65% recall for Diabetes, which is also pretty sparse in our data. Its healthy class recall sits at 62% which is not that bad either. You can clearly see the very visible impact of using class weights in our models. This makes this model the only one to correctly identify the Pre-diabetes, although this model causes the overall accuracy to drop to a 62% but it correctly identifies the Pre-diabetics, with a recall of 26% and the diabetes class with a recall 65%. So obviously these numbers are not very appealing, but this shows the importance of utilizing class weights while training on a severely imbalanced dataset. These results are better given the dataset constraints than the other superficially impressive models.

5.2 Discussion of Model tradeoffs

These results interpret a fundamental issue in medical classification tasks with an imbalanced dataset. Although the High-accuracy model may seem superior but they fail the primary objective of identifying the at risk patients before disease progress. In medical context a model for diabetes screening that never predicts a pre-diabetics, or even diabetes is of no value for preventive healthcare.

The class-weight MLP although sacrifices the accuracy but still achieved more of a balanced results across all the classes. Although it's still imbalanced in terms of our recall scores, this is more of a dataset level constraint at that point. This design choice shows the importance that in reality settings missing a pre-diabetics patient (false-negative) is far more costly than

incorrectly flagging a healthy one for diabetes. The 26% recall results for the pre-diabetics shows a meaningful improvement over other models, which has a 0% recall for pre-diabetes. Also, our diabetes classification performance was all around better, we had a recall of 65% in our diabetes class, while the other models had recalls in the 10-15%. So this is a significant improvement. The only performance loss we see is in the healthy class recall of 62%, while others were in the 90%. This is not that big of an issue as classifying someone as pre-diabetes or even diabetes is not as dangerous to the person's health(if at all) then classifying an actual diabetes patient as healthy.

A self normalizing architecture with SeLU and alpha dropout proved effective. Unlike ReLU that requires batch normalization. SeLU maintains stable activations which in turn enable more consistent training. This model choice proved to be the best to learn representations of the minority classes.

6. Conclusions

In conclusion we have seen the impact of having a highly imbalanced dataset in our machine learning evaluation, namely we train models which only predict the majority class labels and achieve a superficially good score, such as very high accuracies. If you take a look at the per-class recall, this reveals a very stark difference between the different classes, one particular class, the minority class, had a recall value of zero, which means these other models were not even predicting it! This is obviously not at all useful for a clinical context, where

predicting diabetes early in an individual can be the difference between life and death, as when diabetes gets worse, its effects are harder to reverse even with lifestyle changes. Another thing to note is the very low class recall for most of the models for our diabetes class, most of them sat in the 10-15% range, this means that we are not even classifying an individual with diabetes as actually having diabetes. Again this is not useful for a clinical context. Our last most advanced MLP model takes into account the imbalance in the dataset and assigns a higher weight to more sparse classes. This leads to drastic changes in the per-class recall, as all classes are represented. Although class 1 recall, which is our pre-diabetes class remains low, it's still substantially better than the recalls we were getting with the other models, which were all 0, and also substantially better at predicting diabetes in an individual, which is class 2, which in our other more simpler models, were sitting at 10%. Still a problem still remains, we believe we could have extracted more insights from this data, and produced even better models. The problem is that despite our most advanced MLP model being the best performing model overall, it's still a long way to go from achieving clinical usefulness, as in the real world, we need near perfect models to plug our patients' health attributes into. Like getting close to near 100% accuracy, and very high per class recall, for all classes. A model anything lower than this would be dangerous in a medical context.

For the future, we could improve our project, and make its findings more

clinically useful by implementing more aggressive class balancing techniques, one idea that comes to mind is actually manually downscaling the dataset to have similar data instance counts for all three classes. This seems a lot more clinically useful, but there are some downsides to that, such as in the real world, "classes" are not balanced, meaning most people will in fact be healthy, in fact the vast, vast majority will be. Only a minority of patients will have pre-diabetes, or diabetes. This approach still seems to be interesting, and the results might be more insightful than our current results. Also, it allows us to isolate the effect of class imbalance on our models. Another thing we could do is incorporate a time component in our model training, so we ask a question: what if our age progresses by 5 years, and we manage to decrease our BMI, how much does our probability of getting diabetes, or being pre-diabetic change? This would be insightful, also if we incorporate a more time-based health dataset, we could train an LSTM, or even a Transformer to predict if a person will be diabetic, or pre-diabetic. One thing that will make our project results more clinically useful is we grab our advanced MLP architecture probabilities for a given data instance, with proper class balancing of course, and plug it into an LLM, like a Llama model. We also pass into this structured JSON prompt some of the important features, and ask the LLM to give a concise 2-3 sentence advice to the patient, suggesting lifestyle changes, and if risk is elevated for diabetes/pre-diabetes. This LLM response would be a reporting layer, not a predicting layer, predictions would remain with the MLP. It is with these ideas

that we can make machine learning models more useful in critical medical contexts, and ultimately make us prevent diabetes around the world.

7. References

[1] *Maniruzzaman M, Rahman MJ, Ahammed B, Abedin MM. Classification and prediction of diabetes disease using machine learning paradigm. Health Inf Sci Syst. 2020 Jan 3;8(1):7. doi: 10.1007/s13755-019-0095-z. PMID: 31949894; PMCID: PMC6942113.*

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